

Questão 1:

vet = [90, 31, 84, 23, 12, 2, 39, 54, 73, 45]

Merge Sort:

[90, 31, 84, 23, 12 | 2, 39, 54, 73, 45]

[90, 31 | 84, 23, 12]

[2, 39 | 54, 73, 45]

[90 | 31]

[84 | 23, 12]

[2 | 39]

[54 | 73, 45]

[90]

[31]

[84]

[23 | 12]

[2]

[39]

[54]

[73 | 45]

[90]

[31]

[84]

[23]

[12]

[2]

[39]

[54]

[73]

[45]

[31, 90]

[12, 23, 84]

[2, 39]

[45, 54, 73]

[12, 23, 31, 84, 90]

[2, 39, 45, 54, 73]

→ [2, 12, 23, 31, 39, 45, 54, 73, 84, 90]

vetor ordenado

Shell Sort

[90, 31, 84, 23, 12, 2, 39, 54, 73, 45]

tam_vetor = 10

para $s = 1$: $h(s) = 1$

$s = 2$: $2 \cdot h(1) + 1 = 3$

$s = 3$: $2 \cdot 3 + 1 = 7 \leftarrow$ usaremos $h = 7$

$s = 4$: $2 \cdot 7 + 1 = 15$

$i = h$

$[90, 31, 84, 23, 12, 2, 39, 54, 73, 45]$

compara $vet[i]$ com $vet[i-h]$ $54 < 90$

$[54, 31, 84, 23, 12, 2, 39, 90, 73, 45]$ $73 > 31 \rightarrow i++$

$[54, 31, 84, 23, 12, 2, 39, 90, 73, 45]$ $45 < 84$

$[54, 31, 45, 23, 12, 2, 39, 90, 73, 84]$ Final da iteração
de $h = 7$

$i = h = 3$

$[54, 31, 45, 23, 12, 2, 39, 90, 73, 84]$

$[23, 31, 45, 54, 12, 2, 39, 90, 73, 84]$

$[12, 31, 23, 54, 45, 2, 39, 90, 73, 84]$

$[12, 2, 23, 31, 45, 54, 39, 90, 73, 84]$

$[12, 2, 23, 31, 39, 54, 45, 90, 73, 84]$

$[12, 2, 23, 31, 39, 54, 45, 90, 73, 84]$

$[12, 2, 23, 31, 39, 54, 45, 90, 73, 84]$

$[12, 2, 23, 31, 39, 54, 45, 84, 73, 90]$ Final da iteração
 $h = 3$

Quando $h = 1$, se trata somente um
Insertion Sort comum.

$[2, 12, 23, 31, 39, 54, 45, 84, 73, 90]$

$[2, 12, 23, 31, 39, 45, 54, 84, 73, 90]$

$[2, 12, 23, 31, 39, 45, 54, 73, 84, 90]$

Quick sort:

i j p
[90, 31, 84, 23, 12, 2, 39, 54, 73, 45]

i j p
[90, 31, 84, 23, 12, 2, 39, 54, 73, 45]

i j p
[31, 90, 84, 23, 12, 2, 39, 54, 73, 45]

i j p
[31, 90, 84, 23, 12, 2, 39, 54, 73, 45]

i j p
[31, 23, 84, 90, 12, 2, 39, 54, 73, 45]

i j p
[31, 23, 12, 90, 84, 2, 39, 54, 73, 45]

i j p
[31, 23, 12, 2, 84, 90, 39, 54, 73, 45]

i j p
[31, 23, 12, 2, 39, 90, 84, 54, 73, 45]

[31, 23, 12, 2, 39, 45, 84, 54, 73, 90]

i j p
[31, 23, 12, 2, 39] 45 [84, 54, 73, 90]

i j p
[31, 23, 12, 2, 39]

i j p
[31, 23, 12, 2, 39]

i j p
[31, 23, 12, 2, 39]

i j p
[31, 23, 12, 2, 39]

i j p
[31, 23, 12, 2, 39]

i j p
[84, 54, 73, 90]

i j p
[84, 54, 73, 90]

i j p
[84, 54, 73, 90]

i j p
[84, 54, 73, 90]

[[31, 23, 12, 2], 39, 45, [84, 54, 73], 90]

$[31, 23, 12, 21]$
 $[31, 23, 12, 21]$
 $[31, 23, 12, 21]$
 $[31, 23, 12, 21]$
 $[2, 23, 12, 31]$

$[34, 34, 73]$
 $[34, 34, 73]$
 $[34, 34, 73]$
 $[34, 73, 84]$

$[2, [23, 12, 31], 39, 45, 54, 73, 84]$

$[23, 12, 31]$
 $[23, 12, 31]$
 $[23, 12, 31]$

$[2, [23, 12], 31, 39, 45, 54, 73, 84]$

$[2, 12, 23, 31, 39, 45, 54, 73, 84]$

Questão 2

Estruturas de Dados	Inserção			Remoção			Acesso			Memória
	Início	Meio	Fim	Início	Meio	Fim	Início	Meio	Fim	
Lista Vetor	$O(n)$	$O(n)$	$O(1)$	$O(n)$	$O(n)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
Linked List	$O(1)$	$O(n)$	$O(1)$	$O(1)$	$O(n)$	$O(1)$	$O(1)$	$O(n)$	$O(1)$	$O(n)$
Doubly Linked-List	$O(1)$	$O(n)$	$O(1)$	$O(1)$	$O(n)$	$O(1)$	$O(1)$	$O(n)$	$O(1)$	$O(n)$
Pilha vetor	X	X	$O(1)$	X	X	$O(1)$	$O(1)$	$O(1)$	$O(1)$	$O(1)$
Pilha pont	X	X	$O(1)$	X	X	$O(1)$	$O(1)$	$O(n)$	$O(n)$	$O(n)$
Fila vetor	X	X	$O(1)$	$O(1)$	X	X	$O(1)$	$O(1)$	$O(1)$	$O(1)$
Fila pont	X	X	$O(1)$	$O(1)$	X	X	$O(1)$	$O(n)$	$O(n)$	$O(n)$

